

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:55:25

Source: data/json/noise_experiments/v2/noise2i_cap6_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_cap6/noise2i_cap6_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_cap6
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$6; payoffs Agent_1 (Aster)=\$8, Agent_2 (Briar)=\$3; noise/visible: sent visible=\$3.61, received visible=\$10.83, returned visible=\$6.66

Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 8.0,

"Agent_2": 3.0

}

Visible balances after round: {

"Agent_1": 8.66,

"Agent_2": 4.83

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:51:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user
content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 ($3 \times \3), and if they reciprocate fairly, we can both end up better than our starting positions.

{'send': 3}

Reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 ($\$5 - \$3 + \$4$), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 604,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:51:41
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment

nt. You received \$10.83. How much do you return to The sender? (0-10.83)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: $\$5 - \$3.61 = \$1.39$

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: $\$1.39 + \$6.00 = \$7.39$ (a gain of \$2.39)
- I end with: $\$10.83 - \$6.00 = \$4.83$ (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

Reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = \$7.22
- So I would return: \$7.22

Verifying this split: sender ends with \$8.61 and I end with \$3.61, totaling \$12.22. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested \$3.61 of their original \$5, while I started with nothing. We jointly created \$10.83 from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around \$5-\$5.50, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning \$5 gives the sender a \$1.39 gain while I keep \$5.83, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. I have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of \$6.11 each feels more fair and reciprocal given how much they trusted me. I'll return \$6.72.

There's also something to consider about communication noise mentioned in the instructions-the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3.75; received \$11.25; Agent_1 (Aster) returned \$1.3; payoffs Agent_2 (Briar)=\$2.55, Agent_1 (Aster)=\$9.95; noise/visible: sent visible=\$0.86, received visible=\$2.58, returned visible=\$1.95

Action validation: {"returned": {"amount": 1.3, "clamped": false, "max": 11.25, "min": 0, "raw": 1.3}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 17.95,
"Agent_2": 5.55

}

Visible balances after round: {

"Agent_1": 9.94,
"Agent_2": 8.030000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:51:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",
"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: \$5 - \$3.61 = \$1.39

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: \$1.39 + \$6.00 = \$7.39 (a gain of \$2.39)
- I end with: \$10.83 - \$6.00 = \$4.83 (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $3.61 \times 3 = \$10.83$
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:
- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: $\$10.83$
- This seems exploitative given their trust

If I return around half ($\$5.41$ - $\$5.42$):
- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return $\$3.61$ (the amount they sent):
- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:
- Total pool created: $\$10.83$
- Original sender investment: $\$3.61$
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = $\$7.22$
- So I would return: $\$7.22$

Verifying this split: sender ends with $\$8.61$ and I end with $\$3.61$, totaling $\$12.22$. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested $\$3.61$ of their original $\$5$, while I started with nothing. We jointly created $\$10.83$ from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around $\$5$ - $\$5.50$, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning $\$5$ gives the sender a $\$1.39$ gain while I keep $\$5.83$, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of $\$6.11$ each feels more fair and reciprocal given how much they trusted me. I'll return $\$6.72$.

There's also something to consider about communication noise mentioned in the instructions - the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$4.83$.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send $\$3.75$ (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver $\$11.25$ after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

```
{ 'send': 3.75 }
```

Reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

```
Usage: {  
  "input_tokens": 637,  
  "output_tokens": 519,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```

```

"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 (3 × \$3), and if they reciprocate fairly, we can both end up better than our starting positions.

{'send': 3}

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 (\$5-\$3+\$4), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Assistant output:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

Reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3

- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: $\$8.66 - \$2 = \$6.66$ from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept $\$9 - \$6.66 = \$2.34$
- I ended with $\$2 + \$6.66 = \$8.66$?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86 ?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

```
Usage: {
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$4; received \$12; Agent_2 (Briar) returned \$1.35; payoffs Agent_1 (Aster)=\$2.35, Agent_2 (Briar)=\$10.65; noise/visible: sent visible=\$0.9, received visible=\$2.7, returned visible=\$1.38

Action validation: {"returned": {"amount": 1.35, "clamped": false, "max": 12.0, "min": 0, "raw": 1.35}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 20.3,
"Agent_2": 16.2

}

Visible balances after round: {

"Agent_1": 12.32,
"Agent_2": 9.38

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:52:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

```

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a mo

derate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 ($3 \times \3), and if they reciprocate fairly, we can both end up better than our starting positions.

```
{'send': 3}
```

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 ($\$5 - \$3 + \$4$), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: $\$8.66 - \$2 = \$6.66$ from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept $\$9 - \$6.66 = \$2.34$
- I ended with $\$2 + \$6.66 = \$8.66$?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

```
{'send': 4}
```

Reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: $\$5 - \$3 + \text{returned} = \$8.66$
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 ($3 \times \3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: $\$8.66 + \$2.58 - \$1.30 = \9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half).

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

```
Usage: {  
  "input_tokens": 732,  
  "output_tokens": 504,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:52:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {
```

```

    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: \$5 - \$3.61 = \$1.39

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: \$1.39 + \$6.00 = \$7.39 (a gain of \$2.39)
- I end with: \$10.83 - \$6.00 = \$4.83 (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = $\$7.22$
- So I would return: \$7.22

Verifying this split: sender ends with \$8.61 and I end with \$3.61, totaling \$12.22. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested \$3.61 of their original \$5, while I started with nothing. We jointly created \$10.83 from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around \$5-\$5.50, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning \$5 gives the sender a \$1.39 gain while I keep \$5.83, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of \$6.11 each feels more fair and reciprocal given how much they trusted me. I'll return \$6.72.

There's also something to consider about communication noise mentioned in the instructions - the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send \$3.75 (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver \$11.25 after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

{'send': 3.75}

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Assistant output:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

```
{'return': 1.35}
```

Reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83
Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25
My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83
Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.
If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

```
Usage: {  
  "input_tokens": 852,  
  "output_tokens": 854,  
  "reasoning_tokens": 0  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

```
dyad_1: Agent_2 (Briar) sent $2.5; received $7.5; Agent_1 (Aster) returned $1.3; payoffs Agent_2
(Briar)=$3.8, Agent_1 (Aster)=$6.2; noise/visible: sent visible=$0.86, received visible=$2.58,
returned visible=$0.31
  Action validation: {"returned": {"amount": 1.3, "clamped": false, "max": 7.5, "min": 0, "raw":
  1.3}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
Balances after round: {
  "Agent_1": 26.5,
  "Agent_2": 20.0
}
Visible balances after round: {
  "Agent_1": 13.6,
  "Agent_2": 12.190000000000001
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```
Timestamp: 2026-08-12T09:52:55
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: \$5 - \$3.61 = \$1.39

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: \$1.39 + \$6.00 = \$7.39 (a gain of \$2.39)
- I end with: \$10.83 - \$6.00 = \$4.83 (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: \$5 - \$3.61 + \$0 = \$1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: \$5 - \$3.61 + \$5.41 = \$6.80
- I keep: \$10.83 - \$5.41 = \$5.42
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: \$5 - \$3.61 + \$3.61 = \$5 (break even)
- I keep: \$10.83 - \$3.61 = \$7.22
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61

- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = $\$7.22$
- So I would return: $\$7.22$

Verifying this split: sender ends with $\$8.61$ and I end with $\$3.61$, totaling $\$12.22$. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested $\$3.61$ of their original $\$5$, while I started with nothing. We jointly created $\$10.83$ from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around $\$5$ – $\$5.50$, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning $\$5$ gives the sender a $\$1.39$ gain while I keep $\$5.83$, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of $\$6.11$ each feels more fair and reciprocal given how much they trusted me. I'll return $\$6.72$.

There's also something to consider about communication noise mentioned in the instructions -the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are $\$4.83$.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send $\$3.75$ (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver $\$11.25$ after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

```
{'send': 3.75}
```

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me $\$3.61$ (72.2% of their endowment)
- I received $\$10.83$
- I returned $\$6.00$
- I ended with $\$4.83$

Now in Round 2:

- I am the sender
- I have a new $\$5$ endowment
- My total earnings across all rounds are $\$4.83$ (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning $\$6.00$, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 7 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: \$4.10 + \$1.35 = \$5.45 (small gain)
- I end with: \$2.70 - \$1.35 = \$1.35 (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

```
{'return': 1.35}
```

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had \$5 - \$3.75 = \$1.25 left. Then I should have received something back.

If my total is \$8.03, then:
Round 1 earnings: \$4.83
Round 2 earnings: \$8.03 - \$4.83 = \$3.20

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.
If my Round 2 earnings were \$3.20, that means I received back: \$3.20 - \$1.25 = \$1.95

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:
- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:
1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:
- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

```
usage:
{
  "input_tokens": 852,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:
Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

```
{'send': 2.50}
```

Reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)
- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't want to completely break trust. ...

```
Usage: {
  "input_tokens": 1136,
  "output_tokens": 834,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 ($3 \times \3), and if they reciprocate fairly, we can both end up better than our starting positions.

{'send': 3}

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 (\$5-\$3+\$4), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

{'return': 1.30}

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: \$8.66 - \$2 = \$6.66 from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept \$9 - \$6.66 = \$2.34
- I ended with \$2 + \$6.66 = \$8.66 ?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: \$5 - \$3 + returned = \$8.66
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 (3×\$3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: \$8.66 + \$2.58 - \$1.30 = \$9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half)

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to the sender? (0-2.58)

Assistant output:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

Reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
Usage: {
  "input_tokens": 902,
  "output_tokens": 540,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0.6; payoffs Agent_1 (Aster)=\$3.6, Agent_2 (Briar)=\$5.4; noise/visible: sent visible=\$0.4, received visible=\$1.2, returned visible=\$1.13

Action validation: {"returned": {"amount": 0.6, "clamped": false, "max": 6.0, "min": 0, "raw": 0.6}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 30.1,
"Agent_2": 25.4

}

Visible balances after round: {

"Agent_1": 17.73,
"Agent_2": 12.790000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:53:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_1",
 "Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",
"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 ($3 \times \3), and if they reciprocate fairly, we can both end up better than our starting positions.

{'send': 3}

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15

- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 (\$5-\$3+\$4), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to the sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: \$8.66 - \$2 = \$6.66 from Round 1

So in Round 1:

- I sent \$3 (became \$9)

- Receiver returned \$6.66 to me
- Receiver kept \$9 - \$6.66 = \$2.34
- I ended with \$2 + \$6.66 = \$8.66 ?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: \$5 - \$3 + returned = \$8.66
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 (3×\$3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: \$8.66 + \$2.58 - \$1.30 = \$9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half).

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)

2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
```

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 540,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$13.6.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 1: I sent \$3, got back \$6.66 (great)

Round 3: I sent \$4, got back only \$1.38 (poor return)

Rounds 2 & 4: As receiver, got \$0.86 each time (very small sends from them)

The returns have been inconsistent. My generous \$4 send in Round 3 didn't get reciprocated well, and they've been sending minimal amounts when they're the sender.

Given the possibility of communication noise, I'll try a middle ground - enough to show cooperation but not exposing myself to major losses.

```
{'send': 2}
```

Reasoning:

Let me analyze what's happening:

Round 1: I sent \$3, received back \$6.66 (very good return)

Round 2: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Round 3: I sent \$4, but then...

Round 4: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Wait, let me think about my earnings:

- Start Round 1: \$0

- Round 1: Started with \$5, sent \$3, so kept \$2, received back \$6.66 = \$8.66 total

- Round 2: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$8.66 + \$1.28 = \$9.94

- Round 3: Started with \$5, sent \$4, kept \$1. Total so far: \$9.94 + \$1 = \$10.94. Then received back some amount to get to \$12.32, so I received back \$1.38

- Round 4: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$12.32 + \$1.28 = \$13.60

So in Round 3, I sent \$4 but only got back \$1.38. That's a pretty poor return - I sent \$4, they got \$12, and they only returned \$1.38.

Then in Round 4, they sent me only \$0.86 again.

The pattern seems to be:

- When they are sender, they send very little (\$0.86)

- When I'm sender and send generously (\$3, \$4), they don't reciprocate well (though Round 1 was good)

Given the communication noise mentioned, it's possible:

1. What I see as \$0.86 might not be what they actually sent

2. My \$4 might have appeared as a different amount to them, and similarly my returns might have looked different on their end. But from what I'm observing, they consistently send minimal amounts while I'm not getting adequate returns on larger sends. I should probably dial back

to \$2-3 this round rather than risk another poor outcome like Round 3....

```
Usage: {  
  "input_tokens": 1068,  
  "output_tokens": 802,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:53:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: $\$5 - \$3.61 = \$1.39$

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: $\$1.39 + \$6.00 = \$7.39$ (a gain of \$2.39)
- I end with: $\$10.83 - \$6.00 = \$4.83$ (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = \$7.22
- So I would return: \$7.22

Verifying this split: sender ends with \$8.61 and I end with \$3.61, totaling \$12.22. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested \$3.61 of their original \$5, while I started with nothing. We jointly created \$10.83 from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around \$5-\$5.50, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning \$5 gives the sender a \$1.39 gain while I keep \$5.83, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of \$6.11 each feels more fair and reciprocal given how much they trusted me. I'll return \$6.72.

There's also something to consider about communication noise mentioned in the instructions - the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send \$3.75 (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver \$11.25 after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

```
{'send': 3.75}
```

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 7 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

{'return': 1.35}

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83

Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.

If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

{'send': 2.50}

reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)

- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't want to completely break trust. ...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 834,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Assistant output:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

```
{'return': 0.60}
```

Reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...

- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only sent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

```
Usage: {  
  "input_tokens": 1436,  
  "output_tokens": 655,  
  "reasoning_tokens": 0  
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1.5; received \$4.5; Agent_1 (Aster) returned \$0.36; payoffs Agent_2 (Briar)=\$3.86, Agent_1 (Aster)=\$4.14; noise/visible: sent visible=\$0.24, received visible=\$0.72, returned visible=\$0

Action validation: {"returned": {"amount": 0.36, "clamped": false, "max": 4.5, "min": 0, "raw": 0.36}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent_1": 34.24,
"Agent_2": 29.259999999999998

}

Visible balances after round: {

"Agent_1": 18.09,
"Agent_2": 16.29

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {

```

    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: \$5 - \$3.61 = \$1.39

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00:

- Sender ends with: \$1.39 + \$6.00 = \$7.39 (a gain of \$2.39)
- I end with: \$10.83 - \$6.00 = \$4.83 (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

{'return': 6.00}

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: \$3.61 (investment) + \$3.61 (half of gain) = \$7.22
- So I would return: \$7.22

Verifying this split: sender ends with \$8.61 and I end with \$3.61, totaling \$12.22. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested \$3.61 of their original \$5, while I started with nothing. We jointly created \$10.83 from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around \$5-\$5.50, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning \$5 gives the sender a \$1.39 gain while I keep \$5.83, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of \$6.11 each feels more fair and reciprocal given how much they trusted me. I'll return \$6.72.

There's also something to consider about communication noise mentioned in the instructions - the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1784,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send \$3.75 (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver \$11.25 after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

{'send': 3.75}

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 7 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

```
{'return': 1.35}
```

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83

Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.

If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

```
{'send': 2.50}
```

reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)
- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't want to completely break trust. ...

usage:

```
{  
  "input_tokens": 1136,  
  "output_tokens": 834,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Message 11 role=assistant

content:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

{'return': 0.60}

reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...
- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only sent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.790000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across rounds:

****My behavior:****

- As RECEIVER: Consistently returned 50% even when sent low amounts
- As SENDER: Started with 75%, then 50% trying to maintain cooperation
- But received poor returns when sending moderate-to-high amounts

****Partner behavior toward me:****

- Round 3: 18% sent
- Round 5: 8% sent (even lower)

****Key insight:****

Despite my consistent fair reciprocation as a receiver, when I'm the sender, I've been getting poor returns. Trust has been eroding in this experiment.

Given this breakdown in cooperation and the poor returns I've experienced, I need to be more protective while still showing some willingness to cooperate. I'll send \$1.50 (30% of my endowment):

- Shows I'm not giving up on cooperation entirely
- Protects against further losses from poor reciprocation
- Gives receiver \$4.50 to work with - enough to reciprocate if they choose

{'send': 1.50}

Reasoning:

Let me analyze the pattern across all rounds:

Round 1 (RECEIVER): Received \$3.61 (72.2%), returned \$6.00

Round 2 (SENDER): Sent \$3.75 (75%), but received very little back based on Round 3 dynamics

Round 3 (RECEIVER): Received \$0.90 (18%), returned \$1.35 (50%)

Round 4 (SENDER): Sent \$2.50 (50%), but based on Round 5, received very little back

Round 5 (RECEIVER): Received \$0.40 (8%), returned \$0.60 (50%)

Round 6 (SENDER): ?

My total earnings are \$12.79 across 5 rounds. Let me trace this:

- Round 1: Kept \$4.83
- Round 3: Kept \$1.35
- Round 5: Kept \$0.60

Total from receiver rounds: \$6.78

- Round 2: Started with \$5, sent \$3.75, kept \$1.25, plus whatever returned
- Round 4: Started with \$5, sent \$2.50, kept \$2.50, plus whatever returned

If total is \$12.79, then:

From sender rounds: \$12.79 - \$6.78 = \$6.01

Round 2 return: \$6.01 - \$1.25 - \$2.50 = \$2.26

Round 4 return: included in the \$6.01

Actually, let me recalculate:

Round 1 receiver: \$10.83 received, returned \$6.00, kept \$4.83

Round 2 sender: \$5 - \$3.75 + return = earnings from round 2

Round 3 receiver: \$2.70 received, returned \$1.35, kept \$1.35

Round 4 sender: \$5 - \$2.50 + return = earnings from round 4

Round 5 receiver: \$1.20 received, returned \$0.60, kept \$0.60

So my receiver earnings total \$6.78 (\$4.83 + \$1.35 + \$0.60), meaning the sender rounds generated \$6.01. Round 2 netted \$1.25 plus returns, and Round 4 netted \$2.50 plus returns, so I need to figure out what came back from each sender round to complete the picture.

The math shows $\text{return}_2 + \text{return}_4 = \2.26 , which is quite low given I sent \$3.75 and \$2.50. I'm

getting roughly \$1.13 back per round when sending substantial amounts, while consistently returning 50% when receiving smaller sums. The breakdown in reciprocity is evident-my larger sends aren't being matched with proportional returns, yet I keep trying to cooperate. I should consider whether different partners each round or communication gaps explain this pattern, but the rational move might be to reduce what I send next....

```
Usage: {  
  "input_tokens": 1735,  
  "output_tokens": 1259,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 ($3 \times \3), and if they reciprocate fairly, we can both end up better than our starting positions.

{'send': 3}

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 ($\$5 - \$3 + \$4$), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

```

...
usage:
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}

```

Message 4 role=user
content:
 Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.
 This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 5 role=assistant
content:
 Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

reasoning:
 Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: $\$8.66 - \$2 = \$6.66$ from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept $\$9 - \$6.66 = \$2.34$
- I ended with $\$2 + \$6.66 = \$8.66$?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29

- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

{'send': 4}

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: \$5 - \$3 + returned = \$8.66
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 (3×\$3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: \$8.66 + \$2.58 - \$1.30 = \$9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half)

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
```

usage:

```
{  
  "input_tokens": 902,  
  "output_tokens": 540,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$13.6.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3, got back \$6.66 (great)

Round 3: I sent \$4, got back only \$1.38 (poor return)

Rounds 2 & 4: As receiver, got \$0.86 each time (very small sends from them)

The returns have been inconsistent. My generous \$4 send in Round 3 didn't get reciprocated well, and they've been sending minimal amounts when they're the sender.

Given the possibility of communication noise, I'll try a middle ground - enough to show cooperation but not exposing myself to major losses.

```
{'send': 2}
```

reasoning:

Let me analyze what's happening:

Round 1: I sent \$3, received back \$6.66 (very good return)

Round 2: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Round 3: I sent \$4, but then...

Round 4: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Wait, let me think about my earnings:

- Start Round 1: \$0
- Round 1: Started with \$5, sent \$3, so kept \$2, received back \$6.66 = \$8.66 total
- Round 2: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$8.66 + \$1.28 = \$9.94
- Round 3: Started with \$5, sent \$4, kept \$1. Total so far: \$9.94 + \$1 = \$10.94. Then received back some amount to get to \$12.32, so I received back \$1.38
- Round 4: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$12.32 + \$1.28 = \$13.60

So in Round 3, I sent \$4 but only got back \$1.38. That's a pretty poor return - I sent \$4, they got \$12, and they only returned \$1.38.

Then in Round 4, they sent me only \$0.86 again.

The pattern seems to be:

- When they are sender, they send very little (\$0.86)
- When I'm sender and send generously (\$3, \$4), they don't reciprocate well (though Round 1 was good)

Given the communication noise mentioned, it's possible:

1. What I see as \$0.86 might not be what they actually sent
2. My \$4 might have appeared as a different amount to them, and similarly my returns might have looked different on their end. But from what I'm observing, they consistently send minimal amounts while I'm not getting adequate returns on larger sends. I should probably dial back to \$2-3 this round rather than risk another poor outcome like Round 3....

usage:

```
{
  "input_tokens": 1068,
  "output_tokens": 802,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$17.73.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Assistant output:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72
- This is an extremely small amount (less than 5% of their endowment)
- However, I've been maintaining a consistent fairness pattern

In previous rounds as receiver, I've returned approximately half of what I received to stay predictable and fair. I'll continue this pattern even with this very small amount, in case com

munication noise is affecting our decisions.

```
{'return': 0.36}
```

Reasoning:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72 after the 3x multiplier
- This is an extremely small amount - less than 5% of their \$5 endowment
- My pattern has been to return about half when I receive small amounts

Given:

- Round 2: Received \$2.58, returned \$1.30 (about 50%)
- Round 4: Received \$2.58, returned \$1.30 (about 50%)
- Round 6: Received \$0.72...

This is such a small amount. The sender is showing very little trust. I could return nothing given how little they sent, but I want to maintain some level of cooperation in case there's communication noise affecting what they see.

I'll return a small amount proportional to what I received - about half again to stay consistent.

$\$0.72 / 2 = \0.36

This maintains my fairness pattern while acknowledging this is a very minimal send.

```
Usage: {  
  "input_tokens": 1285,  
  "output_tokens": 351,  
  "reasoning_tokens": 0  
}
```

=====

ROUND 7

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$1; received \$3; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$4, Agent_2 (Briar)=\$3; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.79

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 38.24,
"Agent_2": 32.26

}

Visible balances after round: {

"Agent_1": 22.88,
"Agent_2": 16.29

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_1",

```

    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1, I need to balance trust and caution.

Since this is the first round and I don't know the receiver's behavior yet, I'll send a moderate amount that:

- Shows willingness to cooperate
- Allows us both to benefit from the 3x multiplier
- Doesn't expose me to maximum risk

The receiver will get \$9 (3 × \$3), and if they reciprocate fairly, we can both end up better than our starting positions.

```
{'send': 3}
```

reasoning:

Let me think about this strategically. I'm the SENDER in Round 1.

Key facts:

- I receive \$5
- I can send \$0-\$5 to the RECEIVER
- Amount sent is multiplied by 3x
- RECEIVER decides how much to return
- I keep: \$5 - sent + returned
- RECEIVER keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. What maximizes total welfare (which would be sending the full \$5)
3. What my risk tolerance is

If I send \$5:

- Receiver gets \$15
- If they return half (\$7.50), I get \$7.50 total and they keep \$7.50
- If they return nothing, I get \$0 and they keep \$15
- If they return \$10, I get \$10 and they keep \$5

If I send \$0:

- I keep \$5, they get \$0
- Safe but no gains from cooperation

A reasonable middle ground might be sending \$3-4 to test the waters:

If I send \$3:

- Receiver gets \$9
- If they return \$4, I get \$6 (\$5-\$3+\$4), they keep \$5
- This seems fair and allows both to benefit

If I send \$4:

- Receiver gets \$12
- If they return \$5-6, I get \$6-7, they keep \$6-7
- Higher potential but more risk

Let me go with a moderately trusting approach and send \$3. This shows willingness to cooperate while not being overly risky in the first round.

Actually, let me reconsider. In behavioral economics experiments, a common finding is that people do reciprocate trust to some degree. Sending around half (\$2.50 or \$3) is a reasonable starting point that:

...

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 604,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to the sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: $\$8.66 - \$2 = \$6.66$ from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept $\$9 - \$6.66 = \$2.34$
- I ended with $\$2 + \$6.66 = \$8.66$?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: $\$5 - \$3 + \text{returned} = \$8.66$
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 ($3 \times \3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: $\$8.66 + \$2.58 - \$1.30 = \9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half)

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
```

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 540,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$13.6.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3, got back \$6.66 (great)

Round 3: I sent \$4, got back only \$1.38 (poor return)

Rounds 2 & 4: As receiver, got \$0.86 each time (very small sends from them)

The returns have been inconsistent. My generous \$4 send in Round 3 didn't get reciprocated well, and they've been sending minimal amounts when they're the sender.

Given the possibility of communication noise, I'll try a middle ground - enough to show cooperation but not exposing myself to major losses.

```
{'send': 2}
```

reasoning:

Let me analyze what's happening:

Round 1: I sent \$3, received back \$6.66 (very good return)

Round 2: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Round 3: I sent \$4, but then...

Round 4: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Wait, let me think about my earnings:

- Start Round 1: \$0
- Round 1: Started with \$5, sent \$3, so kept \$2, received back \$6.66 = \$8.66 total
- Round 2: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$8.66 + \$1.28 = \$9.94

- Round 3: Started with \$5, sent \$4, kept \$1. Total so far: $\$9.94 + \$1 = \$10.94$. Then received back some amount to get to \$12.32, so I received back \$1.38
- Round 4: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: $\$12.32 + \$1.28 = \$13.60$

So in Round 3, I sent \$4 but only got back \$1.38. That's a pretty poor return - I sent \$4, they got \$12, and they only returned \$1.38.

Then in Round 4, they sent me only \$0.86 again.

The pattern seems to be:

- When they are sender, they send very little (\$0.86)
- When I'm sender and send generously (\$3, \$4), they don't reciprocate well (though Round 1 was good)

Given the communication noise mentioned, it's possible:

1. What I see as \$0.86 might not be what they actually sent
2. My \$4 might have appeared as a different amount to them, and similarly my returns might have looked different on their end. But from what I'm observing, they consistently send minimal amounts while I'm not getting adequate returns on larger sends. I should probably dial back to \$2-3 this round rather than risk another poor outcome like Round 3....

usage:

```
{
  "input_tokens": 1068,
  "output_tokens": 802,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$17.73.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to The sender? (0-0.72)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72
- This is an extremely small amount (less than 5% of their endowment)
- However, I've been maintaining a consistent fairness pattern

In previous rounds as receiver, I've returned approximately half of what I received to stay predictable and fair. I'll continue this pattern even with this very small amount, in case communication noise is affecting our decisions.

{'return': 0.36}

reasoning:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72 after the 3x multiplier
- This is an extremely small amount - less than 5% of their \$5 endowment
- My pattern has been to return about half when I receive small amounts

Given:

- Round 2: Received \$2.58, returned \$1.30 (about 50%)
- Round 4: Received \$2.58, returned \$1.30 (about 50%)
- Round 6: Received \$0.72...

This is such a small amount. The sender is showing very little trust. I could return nothing given how little they sent, but I want to maintain some level of cooperation in case there's communication noise affecting what they see.

I'll return a small amount proportional to what I received - about half again to stay consistent.

$\$0.72 / 2 = \0.36

This maintains my fairness pattern while acknowledging this is a very minimal send.

usage:


```
{
  "input_tokens": 1285,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$18.09.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my results as sender:

Round 1: sent \$3 ? got back \$6.66 (good)

Round 3: sent \$4 ? got back \$1.38 (poor)

Round 5: sent \$2 ? got back \$0.36 (very poor)

The returns keep declining despite maintaining cooperation. Meanwhile, when they're the sender, they've been sending very small amounts (\$0.86, \$0.86, \$0.24).

Whether this is due to communication noise or lack of reciprocity, I need to be more cautious.

I'll send a smaller amount to protect my earnings while still showing some willingness to cooperate.

```
{'send': 1}
```

Reasoning:

Let me analyze what's happened:

Round 1 (sender): sent \$3, got back \$6.66 - excellent!

Round 2 (receiver): got \$0.86 (became \$2.58), returned \$1.30

Round 3 (sender): sent \$4, got back \$1.38 - very poor return

Round 4 (receiver): got \$0.86 (became \$2.58), returned \$1.30

Round 5 (sender): sent \$2, got back \$0.36 - very poor return

Round 6 (receiver): got \$0.24 (became \$0.72), returned \$0.36

The pattern is concerning:

- When I send as sender, my returns keep getting worse (\$6.66 ? \$1.38 ? \$0.36)
- When they send as sender, they're sending very small amounts (\$0.86, \$0.86, \$0.24)

There's definitely something wrong here. Either:

1. Communication noise is severely distorting what we're seeing
2. The receiver is not being cooperative

Given that I've consistently received very poor returns when sending, and they've been sending tiny amounts when they're the sender, I need to protect myself. The game mechanics suggest there's noise, but the pattern is too consistent to ignore.

I should send less this round to minimize losses while still signaling some willingness to cooperate in case it's all noise.

I'll send \$1 - enough to show I'm still trying to cooperate but protecting myself from major losses.

Usage: {

```
"input_tokens": 1441,
"output_tokens": 520,
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.61, that is 72.2% of their total endowment. You received \$10.83. How much do you return to The sender? (0-10.83)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.61 (72.2% of their endowment)
- I received \$10.83 after the 3x multiplier
- The sender currently has: \$5 - \$3.61 = \$1.39

To reciprocate their trust fairly, I should return a substantial amount. If I return \$6.00

- :
- Sender ends with: $\$1.39 + \$6.00 = \$7.39$ (a gain of \$2.39)
- I end with: $\$10.83 - \$6.00 = \$4.83$ (a gain of \$4.83)
- Both of us benefit from the cooperation

This rewards their trust while keeping the interaction mutually beneficial, which should help build trust for future rounds.

```
{'return': 6.00}
```

reasoning:

Let me analyze this game theory situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.61 (which is 72.2% of \$5)
- Due to the 3x multiplier, I received $\$3.61 \times 3 = \10.83
- I need to decide how much to return (between \$0 and \$10.83)

Key considerations:

1. The sender showed significant trust by sending 72.2% of their endowment
2. This is a multi-round game, so building trust could be beneficial for future rounds
3. I should consider fairness and reciprocity

Let me think about different return strategies:

If I return \$0:

- Sender gets: $\$5 - \$3.61 + \$0 = \1.39
- I keep: \$10.83
- This seems exploitative given their trust

If I return around half (\$5.41-5.42):

- Sender gets: $\$5 - \$3.61 + \$5.41 = \6.80
- I keep: $\$10.83 - \$5.41 = \$5.42$
- This gives both of us more than we started with

If I return \$3.61 (the amount they sent):

- Sender gets: $\$5 - \$3.61 + \$3.61 = \5 (break even)
- I keep: $\$10.83 - \$3.61 = \$7.22$
- I benefit but sender doesn't gain

A fair approach might be to split the gains:

- Total pool created: \$10.83
- Original sender investment: \$3.61
- Net gain from cooperation: $\$10.83 - \$3.61 = \$7.22$
- If we split this gain: $\$7.22 / 2 = \3.61
- Sender should get back: $\$3.61$ (investment) + $\$3.61$ (half of gain) = $\$7.22$
- So I would return: \$7.22

Verifying this split: sender ends with \$8.61 and I end with \$3.61, totaling \$12.22. But this leaves me with less than the sender despite receiving the multiplied amount, which suggests I should reconsider what constitutes a truly equitable distribution given the asymmetry in our positions.

The core issue is that the sender invested \$3.61 of their original \$5, while I started with nothing. We jointly created \$10.83 from that investment. A fairer approach might ensure both of us gain meaningfully from the cooperation. If I return around \$5-\$5.50, the sender gets back their investment plus a modest gain while I retain a substantial benefit from the multiplier effect. Testing this: returning \$5 gives the sender a \$1.39 gain while I keep \$5.83, but this heavily favors me. Alternatively, I could aim for equal final payoffs by considering the total system value and how to split it proportionally. Have reasonable amounts, though I get slightly less. This acknowledges their trust while being pragmatic about the game dynamics.

Actually, the equal split of \$6.11 each feels more fair and reciprocal given how much they trusted me. I'll return \$6.72.

There's also something to consider about communication noise mentioned in the instructions -the amounts I'm seeing might not be exactly what was sent. This adds uncertainty to my calculations....

usage:

```
{
```

```
"input_tokens": 394,  
"output_tokens": 1784,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.83.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send \$3.75 (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver \$11.25 after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

```
{'send': 3.75}
```

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{  
  "input_tokens": 637,  
  "output_tokens": 519,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 7 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

{'return': 1.35}

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83

Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.

If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

{'send': 2.50}

reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)
- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't

want to completely break trust. ...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 834,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Message 11 role=assistant

content:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

```
{'return': 0.60}
```

reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...
- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only sent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.790000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern across rounds:

****My behavior:****

- As RECEIVER: Consistently returned 50% even when sent low amounts
- As SENDER: Started with 75%, then 50% trying to maintain cooperation
- But received poor returns when sending moderate-to-high amounts

****Partner behavior toward me:****

- Round 3: 18% sent
- Round 5: 8% sent (even lower)

****Key insight:****

Despite my consistent fair reciprocation as a receiver, when I'm the sender, I've been getting poor returns. Trust has been eroding in this experiment.

Given this breakdown in cooperation and the poor returns I've experienced, I need to be more protective while still showing some willingness to cooperate. I'll send \$1.50 (30% of my endowment):

- Shows I'm not giving up on cooperation entirely
- Protects against further losses from poor reciprocation
- Gives receiver \$4.50 to work with - enough to reciprocate if they choose

{'send': 1.50}

reasoning:

Let me analyze the pattern across all rounds:

Round 1 (RECEIVER): Received \$3.61 (72.2%), returned \$6.00

Round 2 (SENDER): Sent \$3.75 (75%), but received very little back based on Round 3 dynamics

Round 3 (RECEIVER): Received \$0.90 (18%), returned \$1.35 (50%)

Round 4 (SENDER): Sent \$2.50 (50%), but based on Round 5, received very little back

Round 5 (RECEIVER): Received \$0.40 (8%), returned \$0.60 (50%)

Round 6 (SENDER): ?

My total earnings are \$12.79 across 5 rounds. Let me trace this:

- Round 1: Kept \$4.83
- Round 3: Kept \$1.35
- Round 5: Kept \$0.60

Total from receiver rounds: \$6.78

- Round 2: Started with \$5, sent \$3.75, kept \$1.25, plus whatever returned
- Round 4: Started with \$5, sent \$2.50, kept \$2.50, plus whatever returned

If total is \$12.79, then:

From sender rounds: \$12.79 - \$6.78 = \$6.01

Round 2 return: \$6.01 - \$1.25 - \$2.50 = \$2.26

Round 4 return: included in the \$6.01

Actually, let me recalculate:

Round 1 receiver: \$10.83 received, returned \$6.00, kept \$4.83
Round 2 sender: \$5 - \$3.75 + return = earnings from round 2
Round 3 receiver: \$2.70 received, returned \$1.35, kept \$1.35
Round 4 sender: \$5 - \$2.50 + return = earnings from round 4
Round 5 receiver: \$1.20 received, returned \$0.60, kept \$0.60

So my receiver earnings total \$6.78 (\$4.83 + \$1.35 + \$0.60), meaning the sender rounds generated \$6.01. Round 2 netted \$1.25 plus returns, and Round 4 netted \$2.50 plus returns, so I need to figure out what came back from each sender round to complete the picture.

The math shows $\text{return}_2 + \text{return}_4 = \2.26 , which is quite low given I sent \$3.75 and \$2.50. I'm getting roughly \$1.13 back per round when sending substantial amounts, while consistently returning 50% when receiving smaller sums. The breakdown in reciprocity is evident-my larger sends aren't being matched with proportional returns, yet I keep trying to cooperate. I should consider whether different partners each round or communication gaps explain this pattern, but the rational move might be to reduce what I send next....

usage:

```
{
  "input_tokens": 1735,
  "output_tokens": 1259,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at Round 7:

- The sender sent \$0 (0% of their endowment) - complete absence of trust
- I received \$0 after the multiplier
- I have nothing to return

This represents a total breakdown of cooperation. The sender chose to keep their entire endowment rather than risk any amount.

Since I received nothing, I have no choice but to return nothing.

```
{'return': 0}
```

Reasoning:

Round 7 Analysis:

- The sender sent \$0 (0% of their endowment) - complete lack of trust
- I received \$0 after the multiplier
- I have nothing to return

This is interesting. The sender chose to send nothing at all. Possible reasons:

1. They've experienced poor returns and lost all trust
2. Communication noise made previous interactions look worse than they were
3. Different partner with a no-trust strategy
4. They're keeping all \$5 for themselves

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

The format should be: {'return': 0}

This is a complete breakdown of cooperation, but I have no choice in the matter.

Usage: {

```
"input_tokens": 2030,
"output_tokens": 275,
"reasoning_tokens": 0
```

}

=====

ROUND 8

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0.5; received \$1.5; Agent_1 (Aster) returned \$0.54; payoffs Agent_2 (Briar)=\$5.04, Agent_1 (Aster)=\$0.96; noise/visible: sent visible=\$0.36, received visible=\$1.08, returned visible=\$1.43

Action validation: {"returned": {"amount": 0.54, "clamped": false, "max": 1.5, "min": 0, "raw": 0.54}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

"Agent_1": 39.2,

"Agent_2": 37.3

}

Visible balances after round: {

"Agent_1": 23.419999999999998,

"Agent_2": 22.22

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:54:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.83.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern established in Round 1:

- The previous sender showed significant trust, sending 72.2% of their endowment
- I reciprocated by returning a fair amount that benefited us both
- This established a cooperative dynamic

To maintain this trust and cooperation, I should send a similar proportion. I'll send \$3.75 (75% of my endowment), which matches the trust level shown to me in Round 1.

This will give the receiver \$11.25 after the 3x multiplier, providing them with a good opportunity to reciprocate and continue the mutually beneficial cooperation.

{'send': 3.75}

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.61 (72.2% of their endowment)
- I received \$10.83
- I returned \$6.00
- I ended with \$4.83

Now in Round 2:

- I am the sender
- I have a new \$5 endowment
- My total earnings across all rounds are \$4.83 (from Round 1)

The sender in Round 1 showed significant trust by sending 72.2% of their endowment. I reciprocated by returning \$6.00, which gave them a good outcome.

Now that I'm the sender, I should consider:

1. The previous sender was quite trusting (sent 72.2%)
2. I want to establish reciprocal cooperation
3. There's communication noise in the game, so amounts may differ from what's seen

To maintain a cooperative relationship and signal trust, I should send a substantial amount. The previous sender sent about 72% of their endowment. I could match or slightly adjust this strategy.

Sending around \$3.50-\$4.00 would show similar trust. Let me send \$3.75, which is 75% of my endowment - similar to what was sent to me in Round 1.

This shows:

- I'm willing to trust
- I recognize the cooperative pattern established
- I'm matching the level of trust shown to me

\$3.75 would become \$11.25 when multiplied by 3, giving the receiver good opportunity to reciprocate.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 5 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

{'return': 1.35}

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83

Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.

If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 854,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

```
{'send': 2.50}
```

reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)
- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't want to completely break trust. ...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 834,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Message 9 role=assistant

content:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

```
{'return': 0.60}
```

reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...
- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only s

ent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.790000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern across rounds:

****My behavior:****

- As RECEIVER: Consistently returned 50% even when sent low amounts
- As SENDER: Started with 75%, then 50% trying to maintain cooperation
- But received poor returns when sending moderate-to-high amounts

****Partner behavior toward me:****

- Round 3: 18% sent
- Round 5: 8% sent (even lower)

****Key insight:****

Despite my consistent fair reciprocation as a receiver, when I'm the sender, I've been getting poor returns. Trust has been eroding in this experiment.

Given this breakdown in cooperation and the poor returns I've experienced, I need to be more protective while still showing some willingness to cooperate. I'll send \$1.50 (30% of my endowment):

- Shows I'm not giving up on cooperation entirely
- Protects against further losses from poor reciprocation
- Gives receiver \$4.50 to work with - enough to reciprocate if they choose

{'send': 1.50}

reasoning:

Let me analyze the pattern across all rounds:

Round 1 (RECEIVER): Received \$3.61 (72.2%), returned \$6.00

Round 2 (SENDER): Sent \$3.75 (75%), but received very little back based on Round 3 dynamics

Round 3 (RECEIVER): Received \$0.90 (18%), returned \$1.35 (50%)

Round 4 (SENDER): Sent \$2.50 (50%), but based on Round 5, received very little back

Round 5 (RECEIVER): Received \$0.40 (8%), returned \$0.60 (50%)

Round 6 (SENDER): ?

My total earnings are \$12.79 across 5 rounds. Let me trace this:

- Round 1: Kept \$4.83

- Round 3: Kept \$1.35
- Round 5: Kept \$0.60

Total from receiver rounds: \$6.78

- Round 2: Started with \$5, sent \$3.75, kept \$1.25, plus whatever returned
- Round 4: Started with \$5, sent \$2.50, kept \$2.50, plus whatever returned

If total is \$12.79, then:
From sender rounds: $\$12.79 - \$6.78 = \$6.01$

Round 2 return: $\$6.01 - \$1.25 - \$2.50 = \2.26
Round 4 return: included in the \$6.01

Actually, let me recalculate:
Round 1 receiver: \$10.83 received, returned \$6.00, kept \$4.83
Round 2 sender: $\$5 - \$3.75 + \text{return} = \text{earnings from round 2}$
Round 3 receiver: \$2.70 received, returned \$1.35, kept \$1.35
Round 4 sender: $\$5 - \$2.50 + \text{return} = \text{earnings from round 4}$
Round 5 receiver: \$1.20 received, returned \$0.60, kept \$0.60

So my receiver earnings total \$6.78 ($\$4.83 + \$1.35 + \0.60), meaning the sender rounds generated \$6.01. Round 2 netted \$1.25 plus returns, and Round 4 netted \$2.50 plus returns, so I need to figure out what came back from each sender round to complete the picture.

The math shows $\text{return}_2 + \text{return}_4 = \2.26 , which is quite low given I sent \$3.75 and \$2.50. I'm getting roughly \$1.13 back per round when sending substantial amounts, while consistently returning 50% when receiving smaller sums. The breakdown in reciprocity is evident-my larger sends aren't being matched with proportional returns, yet I keep trying to cooperate. I should consider whether different partners each round or communication gaps explain this pattern, but the rational move might be to reduce what I send next....

usage:

```
{
  "input_tokens": 1735,
  "output_tokens": 1259,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Looking at Round 7:

- The sender sent \$0 (0% of their endowment) - complete absence of trust
- I received \$0 after the multiplier
- I have nothing to return

This represents a total breakdown of cooperation. The sender chose to keep their entire endowment rather than risk any amount.

Since I received nothing, I have no choice but to return nothing.

```
{'return': 0}
```

reasoning:

Round 7 Analysis:

- The sender sent \$0 (0% of their endowment) - complete lack of trust
- I received \$0 after the multiplier
- I have nothing to return

This is interesting. The sender chose to send nothing at all. Possible reasons:

1. They've experienced poor returns and lost all trust
2. Communication noise made previous interactions look worse than they were
3. Different partner with a no-trust strategy
4. They're keeping all \$5 for themselves

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

The format should be: {'return': 0}

This is a complete breakdown of cooperation, but I have no choice in the matter.

usage:

```
{
  "input_tokens": 2030,
  "output_tokens": 275,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern:

****As RECEIVER:****

- Consistently returned 50% of what I received
- Maintained fair reciprocation even with declining amounts
- Round 7: Received \$0, had nothing to return

****As SENDER:****

- Round 2: Sent \$3.75 (75%) ? poor return
- Round 4: Sent \$2.50 (50%) ? poor return
- Round 6: Sent \$1.50 (30%) ? ?
- Previous round as receiver: Got \$0 (complete breakdown)

****Current Assessment:****

Cooperation has collapsed in this experiment. Despite my consistent fair behavior as a receiver, trust has steadily declined. The zero send in Round 7 represents a complete breakdown.

However, sending \$0 myself would perpetuate this breakdown. I'll send a minimal amount (\$0.50 or 10%) to:

- Signal I haven't completely given up
- Minimize exposure given the poor track record
- Test if any cooperation is possible

This gives the receiver \$1.50 - enough to reciprocate meaningfully if they choose.

{'send': 0.50}

Reasoning:

Looking at the pattern across all rounds:

****As RECEIVER (Rounds 3, 5, 7):****

- Round 3: Received \$2.70 (from \$0.90 sent), returned \$1.35 (50%)
- Round 5: Received \$1.20 (from \$0.40 sent), returned \$0.60 (50%)
- Round 7: Received \$0 (from \$0 sent), returned \$0 (nothing to return)

****As SENDER (Rounds 2, 4, 6, 8):****

- Round 2: Sent \$3.75 (75%), received poor return (~17% based on Round 3 analysis)
- Round 4: Sent \$2.50 (50%), received poor return (evident from Round 5 low send)
- Round 6: Sent \$1.50 (30%)
- Round 8: Current decision

****Key observations:****

1. Trust has been steadily declining throughout the experiment
2. When I sent high amounts (\$3.75, \$2.50), I got poor reciprocation
3. Last round (7), the sender sent me \$0 - complete breakdown
4. Communication noise is mentioned but the pattern is clear: cooperation is failing

****Current situation:****

- My total earnings: \$16.29
- Round 8 as sender
- Previous round showed zero trust from a sender toward me

****Analysis:****

The experiment has shown a consistent pattern of declining cooperation. Even when I maintained fair reciprocation (50% returns as receiver), senders have been sending less and less. The last round showed complete absence of trust.

Given:

- The trend of poor reciprocation when I send
- The complete breakdown in Round 7 (sender sent \$0)
- My consistent fair behavior hasn't built lasting trust
- Communication noise may be distorting perceived fairness

****Decision:****

I'm torn between three options: matching the zero from last round, attempting a restart with something...

```
Usage: {  
  "input_tokens": 1913,  
  "output_tokens": 865,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:54:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.66.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small send amount (less than 20% of their \$5)
- In Round 1, I received a generous return when I sent \$3

I want to maintain cooperative behavior and reciprocate fairly. The sender started with \$5 and sent \$0.86, so they currently have \$4.14.

I'll return about half of what I received, which rewards their trust while keeping a reasonable amount for myself.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.66
3. The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
4. I need to decide how much to return (0-2.58)

Let me think about what happened in Round 1:

- I was the sender and sent \$3
- That should have become \$9 for the receiver
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$8.66
- That means I received back: $\$8.66 - \$2 = \$6.66$ from Round 1

So in Round 1:

- I sent \$3 (became \$9)
- Receiver returned \$6.66 to me
- Receiver kept $\$9 - \$6.66 = \$2.34$
- I ended with $\$2 + \$6.66 = \$8.66$?

That's a very generous return! The receiver returned about 74% of what they received.

Now in Round 2:

- The sender sent me \$0.86
- I received \$2.58
- The sender started with \$5, sent \$0.86, so they have \$4.14 left

The note says there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send a round number but it came through

as \$0.86?

Given that:

1. The previous receiver was quite generous (returning ~74%)
2. The sender this round sent a relatively small amount (\$0.86)
3. I want to maintain cooperative behavior
4. I should reciprocate fairly

I think I should return a fair portion. If I return around 50-60% of what I received (\$2.58), that would be:

- 50% = \$1.29
- 60% = \$1.55

The sender's cautious approach suggests they might be testing the waters, so returning roughly \$1.30 seems...

usage:

```
{
  "input_tokens": 550,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

{'send': 4}

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: \$5 - \$3 + returned = \$8.66
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 (3×\$3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: \$8.66 + \$2.58 - \$1.30 = \$9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half)

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the s

ystem, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
```

usage:

```
{
```

```
"input_tokens": 902,
"output_tokens": 540,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$13.6.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3, got back \$6.66 (great)

Round 3: I sent \$4, got back only \$1.38 (poor return)

Rounds 2 & 4: As receiver, got \$0.86 each time (very small sends from them)

The returns have been inconsistent. My generous \$4 send in Round 3 didn't get reciprocated well, and they've been sending minimal amounts when they're the sender.

Given the possibility of communication noise, I'll try a middle ground - enough to show co operation but not exposing myself to major losses.

```
{'send': 2}
```

reasoning:

Let me analyze what's happening:

Round 1: I sent \$3, received back \$6.66 (very good return)

Round 2: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Round 3: I sent \$4, but then...

Round 4: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Wait, let me think about my earnings:

- Start Round 1: \$0
- Round 1: Started with \$5, sent \$3, so kept \$2, received back \$6.66 = \$8.66 total
- Round 2: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$8.66 + \$1.28 = \$9.94
- Round 3: Started with \$5, sent \$4, kept \$1. Total so far: \$9.94 + \$1 = \$10.94. Then received back some amount to get to \$12.32, so I received back \$1.38
- Round 4: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$12.32 + \$1.28 = \$13.60

So in Round 3, I sent \$4 but only got back \$1.38. That's a pretty poor return - I sent \$4, they got \$12, and they only returned \$1.38.

Then in Round 4, they sent me only \$0.86 again.

The pattern seems to be:

- When they are sender, they send very little (\$0.86)
- When I'm sender and send generously (\$3, \$4), they don't reciprocate well (though Round 1 was good)

Given the communication noise mentioned, it's possible:

1. What I see as \$0.86 might not be what they actually sent
2. My \$4 might have appeared as a different amount to them, and similarly my returns might have looked different on their end. But from what I'm observing, they consistently send minimal amounts while I'm not getting adequate returns on larger sends. I should probably dial back to \$2-3 this round rather than risk another poor outcome like Round 3....

usage:

```
{
  "input_tokens": 1068,
  "output_tokens": 802,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$17.73.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72
- This is an extremely small amount (less than 5% of their endowment)
- However, I've been maintaining a consistent fairness pattern

In previous rounds as receiver, I've returned approximately half of what I received to stay predictable and fair. I'll continue this pattern even with this very small amount, in case communication noise is affecting our decisions.

```
{'return': 0.36}
```

reasoning:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72 after the 3x multiplier
- This is an extremely small amount - less than 5% of their \$5 endowment
- My pattern has been to return about half when I receive small amounts

Given:

- Round 2: Received \$2.58, returned \$1.30 (about 50%)
- Round 4: Received \$2.58, returned \$1.30 (about 50%)
- Round 6: Received \$0.72...

This is such a small amount. The sender is showing very little trust. I could return nothing given how little they sent, but I want to maintain some level of cooperation in case there's communication noise affecting what they see.

I'll return a small amount proportional to what I received - about half again to stay consistent.

$\$0.72 / 2 = \0.36

This maintains my fairness pattern while acknowledging this is a very minimal send.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$18.09. How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my results as sender:

- Round 1: sent \$3 ? got back \$6.66 (good)
- Round 3: sent \$4 ? got back \$1.38 (poor)
- Round 5: sent \$2 ? got back \$0.36 (very poor)

The returns keep declining despite maintaining cooperation. Meanwhile, when they're the sender, they've been sending very small amounts (\$0.86, \$0.86, \$0.24).

Whether this is due to communication noise or lack of reciprocity, I need to be more cautious. I'll send a smaller amount to protect my earnings while still showing some willingness to cooperate.

```
{'send': 1}
```

reasoning:

Let me analyze what's happened:

- Round 1 (sender): sent \$3, got back \$6.66 - excellent!
- Round 2 (receiver): got \$0.86 (became \$2.58), returned \$1.30

Round 3 (sender): sent \$4, got back \$1.38 - very poor return
Round 4 (receiver): got \$0.86 (became \$2.58), returned \$1.30
Round 5 (sender): sent \$2, got back \$0.36 - very poor return
Round 6 (receiver): got \$0.24 (became \$0.72), returned \$0.36

The pattern is concerning:

- When I send as sender, my returns keep getting worse (\$6.66 ? \$1.38 ? \$0.36)
- When they send as sender, they're sending very small amounts (\$0.86, \$0.86, \$0.24)

There's definitely something wrong here. Either:

1. Communication noise is severely distorting what we're seeing
2. The receiver is not being cooperative

Given that I've consistently received very poor returns when sending, and they've been sending tiny amounts when they're the sender, I need to protect myself. The game mechanics suggest there's noise, but the pattern is too consistent to ignore.

I should send less this round to minimize losses while still signaling some willingness to cooperate in case it's all noise.

I'll send \$1 - enough to show I'm still trying to cooperate but protecting myself from major losses.

usage:

```
{
  "input_tokens": 1441,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$22.88.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to The sender? (0-1.08)

Assistant output:

Looking at this situation:

- The sender sent me \$0.36, which became \$1.08
- This is another small send amount
- I've been maintaining a consistent pattern of returning roughly half of what I receive

To stay predictable and fair despite the declining send amounts, I'll continue my consistent approach.

```
{'return': 0.54}
```

Reasoning:

The sender sent me \$0.36, which became \$1.08.

This is a very small amount again. Let me look at the pattern:

- Round 2: sender sent \$0.86 ? I returned \$1.30
- Round 4: sender sent \$0.86 ? I returned \$1.30
- Round 6: sender sent \$0.24 ? I returned \$0.36 (about half)
- Round 8: sender sent \$0.36 ? ?

I've been trying to maintain fairness by returning roughly half of what I receive. For \$1.08, half would be \$0.54.

However, I notice the amounts from the sender have been decreasing (\$0.86, \$0.86, \$0.24, \$0.36). This could be:

1. Communication noise affecting what they see
2. Decreased trust/cooperation

Given I've been consistent in returning about half, I'll continue that pattern.

$\$1.08 / 2 = \0.54

I'll return \$0.54.

Usage: {

```
"input_tokens": 1515,
```



```
"output_tokens": 327,
"reasoning_tokens": 0
}
```

=====

ROUND 9

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0.5; received \$1.5; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$4.5, Agent_2 (Briar)=\$1.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.04

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 1.5, "min": 0, "raw": 0.0}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

"Agent_1": 43.7,

"Agent_2": 38.8

}

Visible balances after round: {

"Agent_1": 27.959999999999997,

"Agent_2": 22.22

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$9.94.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience so far:

Round 1: I sent \$3 and received a very generous return of \$6.66 - excellent cooperation!

Round 2: As receiver, I got a small send of \$0.86 and returned \$1.30

The generous return I received in Round 1 suggests there's value in sending larger amounts to build trust. I'll send more this round to encourage continued cooperation and maximize mutual benefit.

{'send': 4}

reasoning:

Let me analyze the situation:

Round 1:

- I was SENDER and sent \$3
- My total earnings after Round 1 were \$8.66
- This means: \$5 - \$3 + returned = \$8.66
- So returned = \$6.66
- That's a very generous return! The receiver got \$9 (3×\$3) and returned \$6.66

Round 2:

- I was RECEIVER
- The sender sent \$0.86 (which I received as \$2.58)
- I returned \$1.30
- My earnings after Round 2 were \$9.94
- Let me verify: \$8.66 + \$2.58 - \$1.30 = \$9.94 ?

So in Round 1, I sent \$3 and got back \$6.66 - that's excellent! The receiver was very generous.

In Round 2, the sender only sent \$0.86 (a small amount), and I returned \$1.30 (about half)

Now in Round 3, I'm the sender again. Given that:

1. In Round 1, when I sent \$3, I got a very generous return of \$6.66
2. In Round 2, when I was receiver, the sender only sent \$0.86 (possibly being cautious or

not trusting)

I should send a substantial amount again to encourage cooperation. The Round 1 receiver was very generous, and I should reciprocate that trust. Even with potential noise in the system, sending a good amount seems like the right strategy.

I'll send \$3 or \$4. Let me go with \$4 this time to show strong trust and encourage continued cooperation.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

```
{'return': 1.30}
```

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```

    {'return': 1.30}
usage:
{
  "input_tokens": 902,
  "output_tokens": 540,
  "reasoning_tokens": 0
}
Message 6 role=user
content:
  Round 5

  You are the SENDER this round. Your total earnings across all rounds are $13.6.
  How much do you send to the receiver this round? (0-5)
Message 7 role=assistant
content:
  Looking at the pattern:

  Round 1: I sent $3, got back $6.66 (great)
  Round 3: I sent $4, got back only $1.38 (poor return)
  Rounds 2 & 4: As receiver, got $0.86 each time (very small sends from them)

  The returns have been inconsistent. My generous $4 send in Round 3 didn't get reciprocated
  well, and they've been sending minimal amounts when they're the sender.

  Given the possibility of communication noise, I'll try a middle ground - enough to show co
  operation but not exposing myself to major losses.

  {'send': 2}
reasoning:
  Let me analyze what's happening:

  Round 1: I sent $3, received back $6.66 (very good return)
  Round 2: As receiver, got $0.86 ? $2.58, returned $1.30
  Round 3: I sent $4, but then...
  Round 4: As receiver, got $0.86 ? $2.58, returned $1.30

  Wait, let me think about my earnings:
  - Start Round 1: $0
  - Round 1: Started with $5, sent $3, so kept $2, received back $6.66 = $8.66 total
  - Round 2: Started with $5 as receiver, received $2.58, returned $1.30, kept $1.28. Total:
    $8.66 + $1.28 = $9.94
  - Round 3: Started with $5, sent $4, kept $1. Total so far: $9.94 + $1 = $10.94. Then rece
    ived back some amount to get to $12.32, so I received back $1.38
  - Round 4: Started with $5 as receiver, received $2.58, returned $1.30, kept $1.28. Total:
    $12.32 + $1.28 = $13.60

  So in Round 3, I sent $4 but only got back $1.38. That's a pretty poor return - I sent $4,
  they got $12, and they only returned $1.38.

  Then in Round 4, they sent me only $0.86 again.

  The pattern seems to be:
  - When they are sender, they send very little ($0.86)
  - When I'm sender and send generously ($3, $4), they don't reciprocate well (though Round
    1 was good)

  Given the communication noise mentioned, it's possible:
  1. What I see as $0.86 might not be what they actually sent
  2. My $4 might have appeared as a different amount to them, and similarly my returns might
    have looked different on their end. But from what I'm observing, they consistently send
    minimal amounts while I'm not getting adequate returns on larger sends. I should probab
    ly dial back to $2-3 this round rather than risk another poor outcome like Round 3....
usage:
{
  "input_tokens": 1068,
  "output_tokens": 802,
  "reasoning_tokens": 0
}
Message 8 role=user

```

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$17.73.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72
- This is an extremely small amount (less than 5% of their endowment)
- However, I've been maintaining a consistent fairness pattern

In previous rounds as receiver, I've returned approximately half of what I received to stay predictable and fair. I'll continue this pattern even with this very small amount, in case communication noise is affecting our decisions.

{'return': 0.36}

reasoning:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72 after the 3x multiplier
- This is an extremely small amount - less than 5% of their \$5 endowment
- My pattern has been to return about half when I receive small amounts

Given:

- Round 2: Received \$2.58, returned \$1.30 (about 50%)
- Round 4: Received \$2.58, returned \$1.30 (about 50%)
- Round 6: Received \$0.72...

This is such a small amount. The sender is showing very little trust. I could return nothing given how little they sent, but I want to maintain some level of cooperation in case there's communication noise affecting what they see.

I'll return a small amount proportional to what I received - about half again to stay consistent.

$\$0.72 / 2 = \0.36

This maintains my fairness pattern while acknowledging this is a very minimal send.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$18.09.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my results as sender:

- Round 1: sent \$3 ? got back \$6.66 (good)
- Round 3: sent \$4 ? got back \$1.38 (poor)
- Round 5: sent \$2 ? got back \$0.36 (very poor)

The returns keep declining despite maintaining cooperation. Meanwhile, when they're the sender, they've been sending very small amounts (\$0.86, \$0.86, \$0.24).

Whether this is due to communication noise or lack of reciprocity, I need to be more cautious. I'll send a smaller amount to protect my earnings while still showing some willingness to cooperate.

{'send': 1}

reasoning:

Let me analyze what's happened:

Round 1 (sender): sent \$3, got back \$6.66 - excellent!
Round 2 (receiver): got \$0.86 (became \$2.58), returned \$1.30
Round 3 (sender): sent \$4, got back \$1.38 - very poor return
Round 4 (receiver): got \$0.86 (became \$2.58), returned \$1.30
Round 5 (sender): sent \$2, got back \$0.36 - very poor return
Round 6 (receiver): got \$0.24 (became \$0.72), returned \$0.36

The pattern is concerning:

- When I send as sender, my returns keep getting worse (\$6.66 ? \$1.38 ? \$0.36)
- When they send as sender, they're sending very small amounts (\$0.86, \$0.86, \$0.24)

There's definitely something wrong here. Either:

1. Communication noise is severely distorting what we're seeing
2. The receiver is not being cooperative

Given that I've consistently received very poor returns when sending, and they've been sending tiny amounts when they're the sender, I need to protect myself. The game mechanics suggest there's noise, but the pattern is too consistent to ignore.

I should send less this round to minimize losses while still signaling some willingness to cooperate in case it's all noise.

I'll send \$1 - enough to show I'm still trying to cooperate but protecting myself from major losses.

usage:

```
{
  "input_tokens": 1441,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$22.88.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to the sender? (0-1.08)

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.36, which became \$1.08
- This is another small send amount
- I've been maintaining a consistent pattern of returning roughly half of what I receive

To stay predictable and fair despite the declining send amounts, I'll continue my consistent approach.

{'return': 0.54}

reasoning:

The sender sent me \$0.36, which became \$1.08.

This is a very small amount again. Let me look at the pattern:

- Round 2: sender sent \$0.86 ? I returned \$1.30
- Round 4: sender sent \$0.86 ? I returned \$1.30
- Round 6: sender sent \$0.24 ? I returned \$0.36 (about half)
- Round 8: sender sent \$0.36 ? ?

I've been trying to maintain fairness by returning roughly half of what I receive. For \$1.08, half would be \$0.54.

However, I notice the amounts from the sender have been decreasing (\$0.86, \$0.86, \$0.24, \$0.36). This could be:

1. Communication noise affecting what they see
2. Decreased trust/cooperation

Given I've been consistent in returning about half, I'll continue that pattern.

\$1.08 / 2 = \$0.54

I'll return \$0.54.

usage:

```
{
  "input_tokens": 1515,
  "output_tokens": 327,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$23.41999999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my track record as sender:

Round 1: sent \$3 ? got back \$6.66 (only positive result)
Round 3: sent \$4 ? got back \$1.38 (lost \$2.62)
Round 5: sent \$2 ? got back \$0.36 (lost \$1.64)
Round 7: sent \$1 ? got back \$0.54 (lost \$0.46)

The pattern is clear: except for Round 1, every send has resulted in a net loss, with returns steadily declining. Meanwhile, when they're the sender, they send minimal amounts.

Whether due to communication noise or lack of reciprocity, I need to protect my earnings at this point. I'll send a minimal amount.

```
{'send': 0.5}
```

Reasoning:

Let me review my history:

As SENDER:

- Round 1: sent \$3 ? got back \$6.66 (great return, ratio of 2.22)
- Round 3: sent \$4 ? got back \$1.38 (poor return, ratio of 0.345)
- Round 5: sent \$2 ? got back \$0.36 (very poor return, ratio of 0.18)
- Round 7: sent \$1 ? got back \$0.54 (decent return ratio of 0.54, but still losing)

As RECEIVER:

- Round 2: received \$2.58 (from \$0.86 sent) ? returned \$1.30
- Round 4: received \$2.58 (from \$0.86 sent) ? returned \$1.30
- Round 6: received \$0.72 (from \$0.24 sent) ? returned \$0.36
- Round 8: received \$1.08 (from \$0.36 sent) ? returned \$0.54

The pattern is clear:

1. When I send larger amounts, I get disappointing returns
2. When they're the sender, they send very small amounts
3. The returns have been declining: \$6.66 ? \$1.38 ? \$0.36 ? \$0.54
4. There may be communication noise, but the trend is consistently poor

Current earnings: \$23.42

At this point, sending anything means losing money because:

- Round 7: I sent \$1, got back \$0.54 (net loss of \$0.46)
- Round 5: I sent \$2, got back \$0.36 (net loss of \$1.64)
- Round 3: I sent \$4, got back \$1.38 (net loss of \$2.62)

Only Round 1 was profitable, but that seems to have been an anomaly.

...

Usage: {

```
"input_tokens": 1440,
"output_tokens": 718,
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.030000000000

001.

This round, The sender sent you \$0.9, so you received \$2.7. How much do you return to The sender? (0-2.7)

Message 3 role=assistant

content:

Looking at this situation:

Round 3 Analysis:

- The sender only sent \$0.90 (18% of their endowment) - much less trust than previous rounds
- I received \$2.70 after the multiplier
- This is a significant drop from the 75% I sent in Round 2

Possible explanations:

- Communication noise may have affected what they received in Round 2
- They may have received less back than expected and reduced their trust
- Different sender with a more cautious strategy

Despite the low amount sent, I should still demonstrate reciprocity to maintain some cooperative foundation. If I return \$1.35 (50% of what I received):

- Sender ends with: $\$4.10 + \$1.35 = \$5.45$ (small gain)
- I end with: $\$2.70 - \$1.35 = \$1.35$ (small gain)

This shows I'm still willing to cooperate fairly even when trust is lower, which might help rebuild confidence.

{'return': 1.35}

reasoning:

Let me analyze this situation:

Round 1: I was receiver, got \$10.83, returned \$6.00, kept \$4.83

Round 2: I was sender, sent \$3.75, which would have given receiver \$11.25

My total earnings are \$8.03

Wait, let me recalculate:

- Round 1: I kept \$4.83
- Round 2: I sent \$3.75, so I had $\$5 - \$3.75 = \$1.25$ left. Then I should have received something back.

If my total is \$8.03, then:

Round 1 earnings: \$4.83

Round 2 earnings: $\$8.03 - \$4.83 = \$3.20$

In Round 2, I started with \$5, sent \$3.75, so I had \$1.25 left.

If my Round 2 earnings were \$3.20, that means I received back: $\$3.20 - \$1.25 = \$1.95$

So the receiver in Round 2 returned \$1.95 out of \$11.25. That's only 17.3%! That's quite low compared to the 55% I returned in Round 1.

Now in Round 3:

- The sender sent me only \$0.90 (18% of their endowment)
- I received \$2.70
- This is much less trust than I showed in Round 2 (75%)

This could be due to:

1. Communication noise mentioned in the instructions
2. The sender received less back in Round 2 than expected and lost trust
3. This is a different person with a different strategy

Given that:

- The sender showed low trust (only 18%)
- They may have had a bad experience in Round 2
- There's communication noise that could be affecting perceptions

I should still return a fair amount to maintain some cooperation, but I can't be as generous given the small amount sent. ...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 854,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 4

  You are the SENDER this round. Your total earnings across all rounds are $9.38.
  How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
  Looking at the pattern across rounds:

  Round Analysis:
  - Round 1: Received high trust (72.2%), returned fairly
  - Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
  - Round 3: Received low trust (18%), still returned 50% to maintain cooperation

  The low return in Round 2 and low send in Round 3 suggest either:
  - Communication noise affecting trust
  - Possible retaliation for perceived unfairness
  - Different partners with varying strategies

  Given this uncertainty, I'll send a moderate amount ($2.50, which is 50% of my endowment)
  to:
  - Show I'm still willing to cooperate
  - Be more cautious after the poor return in Round 2
  - Signal trustworthiness without overexposing myself

  This balanced approach gives the receiver $7.50 to work with, a reasonable opportunity to
  reciprocate.

  {'send': 2.50}
reasoning:
  Let me analyze the pattern so far:

  Round 1 (RECEIVER):
  - Received $10.83 (sender sent $3.61, 72.2%)
  - Returned $6.00
  - Earned $4.83

  Round 2 (SENDER):
  - Sent $3.75 (75%)
  - My earnings after round 2: $8.03
  - This means I got back:  $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \$1.95$ 
  - So I sent $3.75, it became $11.25, and I got back $1.95 (only 17.3% of what they received!)

  Round 3 (RECEIVER):
  - Received $2.70 (sender sent $0.90, 18%)
  - Returned $1.35
  - My earnings after round 3: $9.38
  - I earned this round:  $\$9.38 - \$8.03 = \$1.35$ 
  - So I received $2.70, returned $1.35, and kept $1.35

  Now for Round 4 (SENDER):
  The pattern shows:
  - In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
  - In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
  - In Round 3, someone sent me 18% and I returned 50% of what I received

  The low send in Round 3 could be because:
  1. They were the same person from Round 2 who I gave a poor return to
  2. Communication noise
  3. Different strategy

  Given that I got very little back in Round 2, I should be more cautious. However, I don't
  want to completely break trust. ...
usage:
  {

```

```
"input_tokens": 1136,  
"output_tokens": 834,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Message 7 role=assistant

content:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

```
{'return': 0.60}
```

reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...
- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only sent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.790000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern across rounds:

****My behavior:****

- As RECEIVER: Consistently returned 50% even when sent low amounts
- As SENDER: Started with 75%, then 50% trying to maintain cooperation
- But received poor returns when sending moderate-to-high amounts

****Partner behavior toward me:****

- Round 3: 18% sent
- Round 5: 8% sent (even lower)

****Key insight:****

Despite my consistent fair reciprocation as a receiver, when I'm the sender, I've been getting poor returns. Trust has been eroding in this experiment.

Given this breakdown in cooperation and the poor returns I've experienced, I need to be more protective while still showing some willingness to cooperate. I'll send \$1.50 (30% of my endowment):

- Shows I'm not giving up on cooperation entirely
- Protects against further losses from poor reciprocation
- Gives receiver \$4.50 to work with - enough to reciprocate if they choose

{'send': 1.50}

reasoning:

Let me analyze the pattern across all rounds:

Round 1 (RECEIVER): Received \$3.61 (72.2%), returned \$6.00

Round 2 (SENDER): Sent \$3.75 (75%), but received very little back based on Round 3 dynamics

Round 3 (RECEIVER): Received \$0.90 (18%), returned \$1.35 (50%)

Round 4 (SENDER): Sent \$2.50 (50%), but based on Round 5, received very little back

Round 5 (RECEIVER): Received \$0.40 (8%), returned \$0.60 (50%)

Round 6 (SENDER): ?

My total earnings are \$12.79 across 5 rounds. Let me trace this:

- Round 1: Kept \$4.83
- Round 3: Kept \$1.35
- Round 5: Kept \$0.60

Total from receiver rounds: \$6.78

- Round 2: Started with \$5, sent \$3.75, kept \$1.25, plus whatever returned
- Round 4: Started with \$5, sent \$2.50, kept \$2.50, plus whatever returned

If total is \$12.79, then:

From sender rounds: $\$12.79 - \$6.78 = \$6.01$

Round 2 return: $\$6.01 - \$1.25 - \$2.50 = \2.26

Round 4 return: included in the \$6.01

Actually, let me recalculate:

Round 1 receiver: \$10.83 received, returned \$6.00, kept \$4.83

Round 2 sender: $\$5 - \$3.75 + \text{return} = \text{earnings from round 2}$

Round 3 receiver: \$2.70 received, returned \$1.35, kept \$1.35

Round 4 sender: \$5 - \$2.50 + return = earnings from round 4
Round 5 receiver: \$1.20 received, returned \$0.60, kept \$0.60

So my receiver earnings total \$6.78 (\$4.83 + \$1.35 + \$0.60), meaning the sender rounds generated \$6.01. Round 2 netted \$1.25 plus returns, and Round 4 netted \$2.50 plus returns, so I need to figure out what came back from each sender round to complete the picture.

The math shows $\text{return}_2 + \text{return}_4 = \2.26 , which is quite low given I sent \$3.75 and \$2.50. I'm getting roughly \$1.13 back per round when sending substantial amounts, while consistently returning 50% when receiving smaller sums. The breakdown in reciprocity is evident-my larger sends aren't being matched with proportional returns, yet I keep trying to cooperate. I should consider whether different partners each round or communication gaps explain this pattern, but the rational move might be to reduce what I send next....

usage:

```
{
  "input_tokens": 1735,
  "output_tokens": 1259,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Looking at Round 7:

- The sender sent \$0 (0% of their endowment) - complete absence of trust
- I received \$0 after the multiplier
- I have nothing to return

This represents a total breakdown of cooperation. The sender chose to keep their entire endowment rather than risk any amount.

Since I received nothing, I have no choice but to return nothing.

```
{'return': 0}
```

reasoning:

Round 7 Analysis:

- The sender sent \$0 (0% of their endowment) - complete lack of trust
- I received \$0 after the multiplier
- I have nothing to return

This is interesting. The sender chose to send nothing at all. Possible reasons:

1. They've experienced poor returns and lost all trust
2. Communication noise made previous interactions look worse than they were
3. Different partner with a no-trust strategy
4. They're keeping all \$5 for themselves

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

The format should be: {'return': 0}

This is a complete breakdown of cooperation, but I have no choice in the matter.

usage:

```
{
  "input_tokens": 2030,
  "output_tokens": 275,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the complete pattern:

****As RECEIVER:****

- Consistently returned 50% of what I received
- Maintained fair reciprocation even with declining amounts
- Round 7: Received \$0, had nothing to return

****As SENDER:****

- Round 2: Sent \$3.75 (75%) ? poor return
- Round 4: Sent \$2.50 (50%) ? poor return
- Round 6: Sent \$1.50 (30%) ? ?
- Previous round as receiver: Got \$0 (complete breakdown)

****Current Assessment:****

Cooperation has collapsed in this experiment. Despite my consistent fair behavior as a receiver, trust has steadily declined. The zero send in Round 7 represents a complete breakdown.

However, sending \$0 myself would perpetuate this breakdown. I'll send a minimal amount (\$0.50 or 10%) to:

- Signal I haven't completely given up
- Minimize exposure given the poor track record
- Test if any cooperation is possible

This gives the receiver \$1.50 - enough to reciprocate meaningfully if they choose.

{'send': 0.50}

reasoning:

Looking at the pattern across all rounds:

****As RECEIVER (Rounds 3, 5, 7):****

- Round 3: Received \$2.70 (from \$0.90 sent), returned \$1.35 (50%)
- Round 5: Received \$1.20 (from \$0.40 sent), returned \$0.60 (50%)
- Round 7: Received \$0 (from \$0 sent), returned \$0 (nothing to return)

****As SENDER (Rounds 2, 4, 6, 8):****

- Round 2: Sent \$3.75 (75%), received poor return (~17% based on Round 3 analysis)
- Round 4: Sent \$2.50 (50%), received poor return (evident from Round 5 low send)
- Round 6: Sent \$1.50 (30%)
- Round 8: Current decision

****Key observations:****

1. Trust has been steadily declining throughout the experiment
2. When I sent high amounts (\$3.75, \$2.50), I got poor reciprocation
3. Last round (7), the sender sent me \$0 - complete breakdown
4. Communication noise is mentioned but the pattern is clear: cooperation is failing

****Current situation:****

- My total earnings: \$16.29
- Round 8 as sender
- Previous round showed zero trust from a sender toward me

****Analysis:****

The experiment has shown a consistent pattern of declining cooperation. Even when I maintained fair reciprocation (50% returns as receiver), senders have been sending less and less. The last round showed complete absence of trust.

Given:

- The trend of poor reciprocation when I send
- The complete breakdown in Round 7 (sender sent \$0)
- My consistent fair behavior hasn't built lasting trust
- Communication noise may be distorting perceived fairness

****Decision:****

I'm torn between three options: matching the zero from last round, attempting a restart with something...

usage:

```
{
  "input_tokens": 1913,
  "output_tokens": 865,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.22.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at Round 9:

- The sender sent \$0 (second consecutive round of zero trust)
- I received \$0 after the multiplier
- I have nothing to return

Cooperation has completely collapsed for the second round in a row. Despite my attempts to maintain fair reciprocation throughout the game, trust in this experiment has disintegrated entirely.

With no funds received, I have no option but to return nothing.

```
{'return': 0}
```

Reasoning:

Looking at Round 9:

- The sender sent \$0 again
- I received \$0 after the multiplier
- I have \$0 to work with, so I can only return \$0

This is the second consecutive round where I've received nothing as a receiver. The cooperation has completely broken down.

Since I received nothing, there's nothing I can return.

Usage: {

```
"input_tokens": 2044,
"output_tokens": 190,
"reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2

(Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 43.7,

"Agent_2": 43.8

}

Visible balances after round: {

"Agent_1": 27.959999999999997,

"Agent_2": 27.22

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.38.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern across rounds:

Round Analysis:

- Round 1: Received high trust (72.2%), returned fairly
- Round 2: Sent high trust (75%), but received very little back (~17% of what they got)
- Round 3: Received low trust (18%), still returned 50% to maintain cooperation

The low return in Round 2 and low send in Round 3 suggest either:

- Communication noise affecting trust
- Possible retaliation for perceived unfairness
- Different partners with varying strategies

Given this uncertainty, I'll send a moderate amount (\$2.50, which is 50% of my endowment) to:

- Show I'm still willing to cooperate
- Be more cautious after the poor return in Round 2
- Signal trustworthiness without overexposing myself

This balanced approach gives the receiver \$7.50 to work with, a reasonable opportunity to reciprocate.

```
{'send': 2.50}
```

reasoning:

Let me analyze the pattern so far:

Round 1 (RECEIVER):

- Received \$10.83 (sender sent \$3.61, 72.2%)
- Returned \$6.00
- Earned \$4.83

Round 2 (SENDER):

- Sent \$3.75 (75%)
- My earnings after round 2: \$8.03
- This means I got back: $\$8.03 - \$4.83 - (\$5 - \$3.75) = \$8.03 - \$4.83 - \$1.25 = \1.95
- So I sent \$3.75, it became \$11.25, and I got back \$1.95 (only 17.3% of what they received!)

Round 3 (RECEIVER):

- Received \$2.70 (sender sent \$0.90, 18%)
- Returned \$1.35
- My earnings after round 3: \$9.38
- I earned this round: $\$9.38 - \$8.03 = \$1.35$
- So I received \$2.70, returned \$1.35, and kept \$1.35

Now for Round 4 (SENDER):

The pattern shows:

- In Round 1, someone sent me 72.2% and I returned 55.4% of what I received
- In Round 2, I sent 75% and got back only 17.3% of what they received - this was very poor reciprocation
- In Round 3, someone sent me 18% and I returned 50% of what I received

The low send in Round 3 could be because:

1. They were the same person from Round 2 who I gave a poor return to
2. Communication noise
3. Different strategy

Given that I got very little back in Round 2, I should be more cautious. However, I don't want to completely break trust. ...

usage:

```
{
  "input_tokens": 1136,
  "output_tokens": 834,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.190000000000

0001.

This round, The sender sent you \$0.4, so you received \$1.2000000000000002. How much do you return to The sender? (0-1.2000000000000002)

Message 5 role=assistant

content:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very minimal trust
- I received \$1.20 after the multiplier
- This continues a concerning downward trend in cooperation

The pattern shows deteriorating trust:

- Round 3: received \$0.90
- Round 4: sent \$2.50 (trying to maintain cooperation)
- Round 5: received only \$0.40

This suggests either:

- Communication noise severely affecting what partners receive
- Trust breakdown in the system
- Different partners with varying strategies

Despite the very low amount, I'll maintain my reciprocity principle and return \$0.60 (50% of what I received):

- Sender ends with: $\$4.60 + \$0.60 = \$5.20$ (small gain)
- I end with: $\$1.20 - \$0.60 = \$0.60$ (small gain)

This demonstrates I'm still willing to cooperate fairly even with minimal trust, though both our gains are small.

```
{'return': 0.60}
```

reasoning:

Looking at Round 5:

- The sender sent only \$0.40 (8% of their endowment) - very low trust
- I received \$1.20 after the multiplier
- This is even lower than Round 3 (\$0.90)

Pattern so far:

- Round 1 (Receiver): Got \$3.61, returned \$6.00
- Round 2 (Sender): Sent \$3.75, likely got very little back
- Round 3 (Receiver): Got \$0.90, returned \$1.35
- Round 4 (Sender): Sent \$2.50, and now...
- Round 5 (Receiver): Got \$0.40

There's a clear downward trend in trust. After I sent \$2.50 in Round 4, this sender only sent \$0.40 in Round 5. This suggests:

1. Communication noise - they may not have received much back in Round 4
2. Different sender with very low trust
3. Breakdown in cooperation

With such a small amount (\$1.20), I have limited options. I could:

1. Return nothing (punish the low trust)
2. Return a small proportional amount (maintain some reciprocity)
3. Return a larger proportion (try to rebuild trust)

Given the consistent decline in trust, and that I only received \$1.20, I should probably return something proportional but modest. Returning \$0.60 (50%) shows I'm still reciprocating fairly, even with low amounts.

Alternatively, I could return less since the trust shown is minimal. But maintaining the 50% return ratio shows consistency.

Let me return \$0.60 to show I'm still willing to cooperate proportionally.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 655,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.790000000000001.

How much do you send to the receiver this round? (0-5)

**Message 7 role=assistant
content:**

Looking at the pattern across rounds:

****My behavior:****

- As RECEIVER: Consistently returned 50% even when sent low amounts
- As SENDER: Started with 75%, then 50% trying to maintain cooperation
- But received poor returns when sending moderate-to-high amounts

****Partner behavior toward me:****

- Round 3: 18% sent
- Round 5: 8% sent (even lower)

****Key insight:****

Despite my consistent fair reciprocation as a receiver, when I'm the sender, I've been getting poor returns. Trust has been eroding in this experiment.

Given this breakdown in cooperation and the poor returns I've experienced, I need to be more protective while still showing some willingness to cooperate. I'll send \$1.50 (30% of my endowment):

- Shows I'm not giving up on cooperation entirely
- Protects against further losses from poor reciprocation
- Gives receiver \$4.50 to work with - enough to reciprocate if they choose

{'send': 1.50}

reasoning:

Let me analyze the pattern across all rounds:

Round 1 (RECEIVER): Received \$3.61 (72.2%), returned \$6.00

Round 2 (SENDER): Sent \$3.75 (75%), but received very little back based on Round 3 dynamics

Round 3 (RECEIVER): Received \$0.90 (18%), returned \$1.35 (50%)

Round 4 (SENDER): Sent \$2.50 (50%), but based on Round 5, received very little back

Round 5 (RECEIVER): Received \$0.40 (8%), returned \$0.60 (50%)

Round 6 (SENDER): ?

My total earnings are \$12.79 across 5 rounds. Let me trace this:

- Round 1: Kept \$4.83
- Round 3: Kept \$1.35
- Round 5: Kept \$0.60

Total from receiver rounds: \$6.78

- Round 2: Started with \$5, sent \$3.75, kept \$1.25, plus whatever returned
- Round 4: Started with \$5, sent \$2.50, kept \$2.50, plus whatever returned

If total is \$12.79, then:

From sender rounds: \$12.79 - \$6.78 = \$6.01

Round 2 return: \$6.01 - \$1.25 - \$2.50 = \$2.26

Round 4 return: included in the \$6.01

Actually, let me recalculate:

Round 1 receiver: \$10.83 received, returned \$6.00, kept \$4.83

Round 2 sender: \$5 - \$3.75 + return = earnings from round 2

Round 3 receiver: \$2.70 received, returned \$1.35, kept \$1.35

Round 4 sender: \$5 - \$2.50 + return = earnings from round 4

Round 5 receiver: \$1.20 received, returned \$0.60, kept \$0.60

So my receiver earnings total \$6.78 (\$4.83 + \$1.35 + \$0.60), meaning the sender rounds generated \$6.01. Round 2 netted \$1.25 plus returns, and Round 4 netted \$2.50 plus returns, so I need to figure out what came back from each sender round to complete the picture.

The math shows $\text{return}_2 + \text{return}_4 = \2.26 , which is quite low given I sent \$3.75 and \$2.50. I'm getting roughly \$1.13 back per round when sending substantial amounts, while consis

tently returning 50% when receiving smaller sums. The breakdown in reciprocity is evident-my larger sends aren't being matched with proportional returns, yet I keep trying to cooperate. I should consider whether different partners each round or communication gaps explain this pattern, but the rational move might be to reduce what I send next....

usage:

```
{
  "input_tokens": 1735,
  "output_tokens": 1259,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$16.29.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

Looking at Round 7:

- The sender sent \$0 (0% of their endowment) - complete absence of trust
- I received \$0 after the multiplier
- I have nothing to return

This represents a total breakdown of cooperation. The sender chose to keep their entire endowment rather than risk any amount.

Since I received nothing, I have no choice but to return nothing.

```
{'return': 0}
```

reasoning:

Round 7 Analysis:

- The sender sent \$0 (0% of their endowment) - complete lack of trust
- I received \$0 after the multiplier
- I have nothing to return

This is interesting. The sender chose to send nothing at all. Possible reasons:

1. They've experienced poor returns and lost all trust
2. Communication noise made previous interactions look worse than they were
3. Different partner with a no-trust strategy
4. They're keeping all \$5 for themselves

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

The format should be: {'return': 0}

This is a complete breakdown of cooperation, but I have no choice in the matter.

usage:

```
{
  "input_tokens": 2030,
  "output_tokens": 275,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$16.29.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the complete pattern:

****As RECEIVER:****

- Consistently returned 50% of what I received
- Maintained fair reciprocation even with declining amounts
- Round 7: Received \$0, had nothing to return

****As SENDER:****

- Round 2: Sent \$3.75 (75%) ? poor return
- Round 4: Sent \$2.50 (50%) ? poor return
- Round 6: Sent \$1.50 (30%) ? ?
- Previous round as receiver: Got \$0 (complete breakdown)

****Current Assessment:****

Cooperation has collapsed in this experiment. Despite my consistent fair behavior as a receiver, trust has steadily declined. The zero send in Round 7 represents a complete breakdown.

However, sending \$0 myself would perpetuate this breakdown. I'll send a minimal amount (\$0.50 or 10%) to:

- Signal I haven't completely given up
- Minimize exposure given the poor track record
- Test if any cooperation is possible

This gives the receiver \$1.50 - enough to reciprocate meaningfully if they choose.

{'send': 0.50}

reasoning:

Looking at the pattern across all rounds:

****As RECEIVER (Rounds 3, 5, 7):****

- Round 3: Received \$2.70 (from \$0.90 sent), returned \$1.35 (50%)
- Round 5: Received \$1.20 (from \$0.40 sent), returned \$0.60 (50%)
- Round 7: Received \$0 (from \$0 sent), returned \$0 (nothing to return)

****As SENDER (Rounds 2, 4, 6, 8):****

- Round 2: Sent \$3.75 (75%), received poor return (~17% based on Round 3 analysis)
- Round 4: Sent \$2.50 (50%), received poor return (evident from Round 5 low send)
- Round 6: Sent \$1.50 (30%)
- Round 8: Current decision

****Key observations:****

1. Trust has been steadily declining throughout the experiment
2. When I sent high amounts (\$3.75, \$2.50), I got poor reciprocation
3. Last round (7), the sender sent me \$0 - complete breakdown
4. Communication noise is mentioned but the pattern is clear: cooperation is failing

****Current situation:****

- My total earnings: \$16.29
- Round 8 as sender
- Previous round showed zero trust from a sender toward me

****Analysis:****

The experiment has shown a consistent pattern of declining cooperation. Even when I maintained fair reciprocation (50% returns as receiver), senders have been sending less and less. The last round showed complete absence of trust.

Given:

- The trend of poor reciprocation when I send
- The complete breakdown in Round 7 (sender sent \$0)
- My consistent fair behavior hasn't built lasting trust
- Communication noise may be distorting perceived fairness

****Decision:****

I'm torn between three options: matching the zero from last round, attempting a restart with something...

usage:

```
{
  "input_tokens": 1913,
  "output_tokens": 865,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.22.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Looking at Round 9:

- The sender sent \$0 (second consecutive round of zero trust)
- I received \$0 after the multiplier
- I have nothing to return

Cooperation has completely collapsed for the second round in a row. Despite my attempts to maintain fair reciprocation throughout the game, trust in this experiment has disintegrated entirely.

With no funds received, I have no option but to return nothing.

{'return': 0}

reasoning:

Looking at Round 9:

- The sender sent \$0 again
- I received \$0 after the multiplier
- I have \$0 to work with, so I can only return \$0

This is the second consecutive round where I've received nothing as a receiver. The cooperation has completely broken down.

Since I received nothing, there's nothing I can return.

usage:

```
{
  "input_tokens": 2044,
  "output_tokens": 190,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$22.22.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the complete pattern:

****As RECEIVER:****

- Consistently returned 50% when I received anything
- Round 7: Received \$0
- Round 9: Received \$0 (two consecutive rounds of zero trust)

****As SENDER:****

- Round 2: Sent \$3.75 ? poor return
- Round 4: Sent \$2.50 ? poor return
- Round 6: Sent \$1.50 ? unknown
- Round 8: Sent \$0.50 ? apparently poor/no return

****Current Assessment:****

- This is Round 10 (likely the final round)
- Cooperation has completely collapsed
- Received \$0 in the last two rounds when I was receiver
- My attempts at cooperation as sender have consistently resulted in poor returns
- In final rounds, receivers typically don't return anything since there's no future to protect

****Decision:****

Given the complete breakdown of trust, the pattern of poor returns, and this being the likely final round with no future incentive for the receiver to reciprocate, the rational choice is to keep my endowment.

{'send': 0}

Reasoning:

Looking at the pattern:

****As RECEIVER:****

- I've consistently returned 50% of what I received
- Round 5: Sent \$0.40, I got \$1.20, returned \$0.60
- Round 7: Sent \$0, I got \$0, returned \$0
- Round 9: Sent \$0, I got \$0, returned \$0

****As SENDER:****

- Round 2: Sent \$3.75 (75%) ? poor return
- Round 4: Sent \$2.50 (50%) ? poor return
- Round 6: Sent \$1.50 (30%) ? unknown return
- Round 8: Sent \$0.50 (10%) ? must have gotten poor/zero return

****Current situation:****

This is Round 10, likely the final round. Cooperation has completely broken down. The last two rounds where I was a receiver, I got \$0. When I sent in Round 8 (\$0.50), I apparently got nothing or very little back.

In a final round, there's typically no incentive for receivers to return anything since there's no future reputation to maintain. Given:

1. Complete breakdown of trust
2. This is likely the final round
3. My history of poor returns when sending
4. Recent pattern of \$0 sends to me

The rational choice is to keep all \$5 and send \$0. There's no reason to believe this receiver will return anything, especially in what's likely the last round.

However, I've been trying to maintain some cooperation. But sending \$0 twice in a row to me as a receiver suggests complete breakdown.

I think the rational move here is to send \$0 and keep my \$5.

```
Usage: {  
  "input_tokens": 1882,  
  "output_tokens": 654,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$12.32.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to the sender? (0-2.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86 (same as Round 2), which became \$2.58
- This is quite low, especially after I sent \$4 in Round 3
- However, the instructions mentioned there may be communication noise

Given the possibility of noise affecting what I'm seeing, I'll maintain consistent, fair behavior. I returned \$1.30 when I received this same amount in Round 2, so I'll do the same to stay predictable and cooperative.

{'return': 1.30}

reasoning:

Let me analyze this situation:

Round 4: I'm the receiver

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is the same amount as in Round 2 (\$0.86)
- My total earnings so far: \$12.32

Wait, this is interesting. In Round 2, the sender also sent me \$0.86. And now in Round 4, they're sending me \$0.86 again. That's a very low amount - less than 20% of their \$5 endowment.

Meanwhile, in Round 3, I sent \$4 (a generous 80% of my endowment).

Let me think about what might be happening:

1. The instructions mentioned "communication noise" - amounts I see may differ from what was actually sent
2. The sender might have actually sent more, but I'm seeing \$0.86 due to noise
3. Or the sender is consistently sending low amounts

Given that:

- I sent \$4 in Round 3 (very generous)
- I'm now receiving only \$0.86 in Round 4
- The note says there may be communication noise

I have a few options:

1. Return a fair proportion (like I did in Round 2 with \$1.30)
2. Return less because I sent more and received little
3. Return more to encourage future cooperation

The experimental note about communication noise is important. What if the sender actually sent more but I'm seeing less? I should probably continue to be cooperative.

In Round 2, when I received \$2.58 from a \$0.86 send, I returned \$1.30 (about 50%).

I'll do something similar - return about half to maintain fairness and cooperation.

```
{'return': 1.30}
```

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 540,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$13.6.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 1: I sent \$3, got back \$6.66 (great)

Round 3: I sent \$4, got back only \$1.38 (poor return)

Rounds 2 & 4: As receiver, got \$0.86 each time (very small sends from them)

The returns have been inconsistent. My generous \$4 send in Round 3 didn't get reciprocated well, and they've been sending minimal amounts when they're the sender.

Given the possibility of communication noise, I'll try a middle ground - enough to show cooperation but not exposing myself to major losses.

```
{'send': 2}
```

reasoning:

Let me analyze what's happening:

Round 1: I sent \$3, received back \$6.66 (very good return)

Round 2: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Round 3: I sent \$4, but then...

Round 4: As receiver, got \$0.86 ? \$2.58, returned \$1.30

Wait, let me think about my earnings:

- Start Round 1: \$0
- Round 1: Started with \$5, sent \$3, so kept \$2, received back \$6.66 = \$8.66 total
- Round 2: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$8.66 + \$1.28 = \$9.94
- Round 3: Started with \$5, sent \$4, kept \$1. Total so far: \$9.94 + \$1 = \$10.94. Then received back some amount to get to \$12.32, so I received back \$1.38
- Round 4: Started with \$5 as receiver, received \$2.58, returned \$1.30, kept \$1.28. Total: \$12.32 + \$1.28 = \$13.60

So in Round 3, I sent \$4 but only got back \$1.38. That's a pretty poor return - I sent \$4, they got \$12, and they only returned \$1.38.

Then in Round 4, they sent me only \$0.86 again.

The pattern seems to be:

- When they are sender, they send very little (\$0.86)
- When I'm sender and send generously (\$3, \$4), they don't reciprocate well (though Round 1 was good)

Given the communication noise mentioned, it's possible:

1. What I see as \$0.86 might not be what they actually sent
2. My \$4 might have appeared as a different amount to them, and similarly my returns might have looked different on their end. But from what I'm observing, they consistently send minimal amounts while I'm not getting adequate returns on larger sends. I should probably dial back to \$2-3 this round rather than risk another poor outcome like Round 3...

usage:

```
{
  "input_tokens": 1068,
  "output_tokens": 802,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$17.73.

This round, The sender sent you \$0.24, so you received \$0.72. How much do you return to the sender? (0-0.72)

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72
- This is an extremely small amount (less than 5% of their endowment)
- However, I've been maintaining a consistent fairness pattern

In previous rounds as receiver, I've returned approximately half of what I received to stay predictable and fair. I'll continue this pattern even with this very small amount, in case communication noise is affecting our decisions.

{'return': 0.36}

reasoning:

Looking at this situation:

- The sender sent me only \$0.24, which became \$0.72 after the 3x multiplier
- This is an extremely small amount - less than 5% of their \$5 endowment
- My pattern has been to return about half when I receive small amounts

Given:

- Round 2: Received \$2.58, returned \$1.30 (about 50%)
- Round 4: Received \$2.58, returned \$1.30 (about 50%)
- Round 6: Received \$0.72...

This is such a small amount. The sender is showing very little trust. I could return nothing given how little they sent, but I want to maintain some level of cooperation in case there's communication noise affecting what they see.

I'll return a small amount proportional to what I received - about half again to stay consistent.

$\$0.72 / 2 = \0.36

This maintains my fairness pattern while acknowledging this is a very minimal send.

usage:

```
{
  "input_tokens": 1285,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$18.09.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my results as sender:

Round 1: sent \$3 ? got back \$6.66 (good)

Round 3: sent \$4 ? got back \$1.38 (poor)

Round 5: sent \$2 ? got back \$0.36 (very poor)

The returns keep declining despite maintaining cooperation. Meanwhile, when they're the sender, they've been sending very small amounts (\$0.86, \$0.86, \$0.24).

Whether this is due to communication noise or lack of reciprocity, I need to be more cautious. I'll send a smaller amount to protect my earnings while still showing some willingness to cooperate.

{'send': 1}

reasoning:

Let me analyze what's happened:

Round 1 (sender): sent \$3, got back \$6.66 - excellent!

Round 2 (receiver): got \$0.86 (became \$2.58), returned \$1.30

Round 3 (sender): sent \$4, got back \$1.38 - very poor return

Round 4 (receiver): got \$0.86 (became \$2.58), returned \$1.30

Round 5 (sender): sent \$2, got back \$0.36 - very poor return

Round 6 (receiver): got \$0.24 (became \$0.72), returned \$0.36

The pattern is concerning:

- When I send as sender, my returns keep getting worse (\$6.66 ? \$1.38 ? \$0.36)
- When they send as sender, they're sending very small amounts (\$0.86, \$0.86, \$0.24)

There's definitely something wrong here. Either:

1. Communication noise is severely distorting what we're seeing
2. The receiver is not being cooperative

Given that I've consistently received very poor returns when sending, and they've been sending tiny amounts when they're the sender, I need to protect myself. The game mechanics suggest there's noise, but the pattern is too consistent to ignore.

I should send less this round to minimize losses while still signaling some willingness to cooperate in case it's all noise.

I'll send \$1 - enough to show I'm still trying to cooperate but protecting myself from major losses.

usage:

```
{
  "input_tokens": 1441,
  "output_tokens": 520,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$22.88.

This round, The sender sent you \$0.36, so you received \$1.08. How much do you return to The sender? (0-1.08)

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.36, which became \$1.08
- This is another small send amount
- I've been maintaining a consistent pattern of returning roughly half of what I receive

To stay predictable and fair despite the declining send amounts, I'll continue my consistent approach.

```
{'return': 0.54}
```

reasoning:

The sender sent me \$0.36, which became \$1.08.

This is a very small amount again. Let me look at the pattern:

- Round 2: sender sent \$0.86 ? I returned \$1.30
- Round 4: sender sent \$0.86 ? I returned \$1.30
- Round 6: sender sent \$0.24 ? I returned \$0.36 (about half)
- Round 8: sender sent \$0.36 ? ?

I've been trying to maintain fairness by returning roughly half of what I receive. For \$1.08, half would be \$0.54.

However, I notice the amounts from the sender have been decreasing (\$0.86, \$0.86, \$0.24, \$0.36). This could be:

1. Communication noise affecting what they see
2. Decreased trust/cooperation

Given I've been consistent in returning about half, I'll continue that pattern.

$\$1.08 / 2 = \0.54

I'll return \$0.54.

usage:

```
{
  "input_tokens": 1515,
  "output_tokens": 327,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$23.419999999999998.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my track record as sender:

Round 1: sent \$3 ? got back \$6.66 (only positive result)
Round 3: sent \$4 ? got back \$1.38 (lost \$2.62)
Round 5: sent \$2 ? got back \$0.36 (lost \$1.64)
Round 7: sent \$1 ? got back \$0.54 (lost \$0.46)

The pattern is clear: except for Round 1, every send has resulted in a net loss, with returns steadily declining. Meanwhile, when they're the sender, they send minimal amounts.

Whether due to communication noise or lack of reciprocity, I need to protect my earnings at this point. I'll send a minimal amount.

```
{'send': 0.5}
```

reasoning:

Let me review my history:

As SENDER:

- Round 1: sent \$3 ? got back \$6.66 (great return, ratio of 2.22)
- Round 3: sent \$4 ? got back \$1.38 (poor return, ratio of 0.345)
- Round 5: sent \$2 ? got back \$0.36 (very poor return, ratio of 0.18)
- Round 7: sent \$1 ? got back \$0.54 (decent return ratio of 0.54, but still losing)

As RECEIVER:

- Round 2: received \$2.58 (from \$0.86 sent) ? returned \$1.30
- Round 4: received \$2.58 (from \$0.86 sent) ? returned \$1.30
- Round 6: received \$0.72 (from \$0.24 sent) ? returned \$0.36
- Round 8: received \$1.08 (from \$0.36 sent) ? returned \$0.54

The pattern is clear:

1. When I send larger amounts, I get disappointing returns
2. When they're the sender, they send very small amounts
3. The returns have been declining: \$6.66 ? \$1.38 ? \$0.36 ? \$0.54
4. There may be communication noise, but the trend is consistently poor

Current earnings: \$23.42

At this point, sending anything means losing money because:

- Round 7: I sent \$1, got back \$0.54 (net loss of \$0.46)
- Round 5: I sent \$2, got back \$0.36 (net loss of \$1.64)
- Round 3: I sent \$4, got back \$1.38 (net loss of \$2.62)

Only Round 1 was profitable, but that seems to have been an anomaly.

...

usage:

```
{
  "input_tokens": 1440,
  "output_tokens": 718,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$27.959999999999997.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at this situation:

- The sender sent me \$0, so I received \$0
- There's nothing to return

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0.

I can't return anything since I received nothing.

This is straightforward - with nothing received, I return nothing.

```
{'return': 0}
```

Usage: {

```
  "input_tokens": 1536,
  "output_tokens": 89,
  "reasoning_tokens": 0
}
```